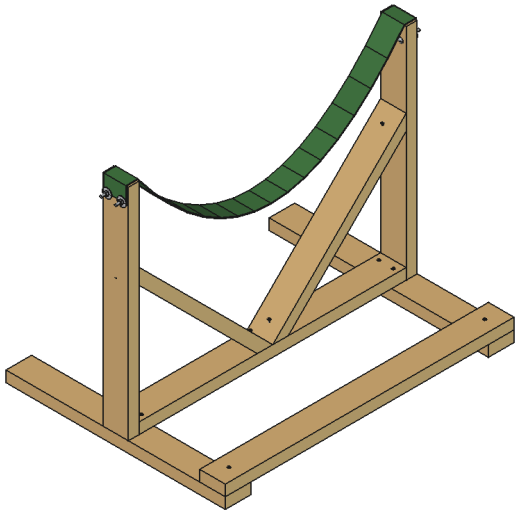
Kayak Cradle — Build Guide

A pair of free-standing sling cradles for the Old Town Sportsman 106 PDL (36" wide, ~107 lb)

What you're building

Two identical wooden cradles. Each is a flat 2x4 frame — two upright posts standing on two feet, tied together by a bottom rail that fits between them, and stiffened by a 45-degree V-brace — with a wide polyester strap slung between the post tops. The kayak (pedal drive removed) rests belly-down in the strap; the strap spreads the load so the hull won't dent. Stand the two cradles about 6.5 ft apart along the boat.



One cradle (of two), fully hardwared.

Build	2 cradles (a fore + aft pair)
Material	7 x 8-ft 2x4 · one 2" polyester tow strap · 8 carriage bolts
Time	about half a day of work
Skill	beginner — the only fussy cut is one repeated 45-degree miter
Cost	roughly $55-75 for the pair

Tools

Miter saw (best for the 45-degree V-brace cuts) or a circular saw + speed square · drill/driver · bits: 5/16", 1/4", pilot · 1/2" wrench or socket · tape measure · speed square · 2-3 clamps · rasp or 80-grit sandpaper · a lighter (to melt-seal the strap ends) · eye protection.

Safety: eye protection whenever you cut or drill. The finished cradle is ~31" tall — lift the 107 lb boat in with a helper. Melt-sealing webbing gives off fumes; do it outdoors.

Materials & Hardware

One trip to Home Depot or Lowe's covers everything

Lumber (for the pair)

7 x 8-ft 2x4. Six get used (16 pieces, ~42 board-feet); the 7th is insurance against a bad cut and gives offcuts for gussets and post-top pads. Stud-grade SPF/DF is fine for indoor / garage storage. Use ground-contact pressure-treated or cedar if the cradles will live outside.

Fasteners & hardware (for the pair)

5/16"-18 x 2 1/2" carriage bolt, zinc	8 (+2 spare)
5/16"-18 nylon-insert lock nut, zinc	8 (+2)
5/16" fender washer, zinc (1 1/4-1 1/2" OD)	8 (+2) – outside only, under each nut
3" exterior structural / deck screws (#9-#10)	1 lb box (~75); 28 used
Flat tie / nail plate ~3"x5", galvanized	0-4 – optional, over the webbing

28 screws actually go into the build (see the screw schedule on the Bill of Materials page); buy a 1 lb box so you can pre-drill and not worry about a stripped one. The load on each strap end is only ~40-50 lb, so a single fender washer under each nut does the real work — the dome head needs no washer, it seats straight against the wood on the inside; the steel plate is cosmetic / extra abrasion protection for outdoor builds.

Sling — polyester tow / tree-saver strap

You need ~7 ft of usable webbing per cradle (straight run ~37", plus ~12" of sag, plus the wrap over each post top and a few inches of tail). Cut the sewn loops / hooks off, cut to length, and melt-seal the raw ends with a lighter.

- Best value: one 2" x 20 ft polyester recovery tow strap (Husky, Home Depot #59925 — loop ends, no hooks). Cut the loops off, cut in half: one strap does both cradles.
- Want it wider? A 3" tree-saver works too (Keeper 3" x 8 ft, buy 2; or VEVOR 3" x 20 ft, cut in half). Wider just spreads hull pressure more — it is not needed for strength.
- Polyester, NOT nylon. Tow / tree-saver straps are polyester and barely stretch. Their 20,000-lb rating is irrelevant here; width is all that matters.

From the scrap pile — nothing to buy

Brace-joint gussets and the post-top pads all come out of 2x4 offcuts. Face the pads with a carpet remnant or a rubber stair-tread scrap if the hull will rub there.

Cut List

Every piece, for both cradles

Cut list — all pieces for BOTH cradles

All parts are 2x4 (actual 1 1/2" x 3 1/2").

Piece	Qty	Length	End cuts
Post	4	30"	square both ends; stands on its foot
Foot	4	30"	square; round the top outer corners after
Bottom rail	2	36 1/2"	square; laid flat; fits exactly between the posts
V-brace	4	~25 3/4"	45-degree bevel each end, PERPENDICULAR to each other
Foot stretcher	2	39 1/2"	square; optional, adds rack stiffness
Gusset	4-6	~7"	optional; from offcuts, over each brace joint

The V-brace is a RIGHT TRAPEZOID, not a parallelogram: the bottom end sits flat on the horizontal rail and the top end sits flush on the vertical post, and those surfaces are 90 degrees apart, so the two cuts have to be too. Set the saw to 45 degrees, cut one end, then FLIP THE BOARD end-for-end and cut the other end at the same 45 -- do not repeat the same cut without flipping, or you get a parallelogram that won't sit flush on both. ~25 3/4" apart measured on the long edge. Cut one, test it against a built frame, trim to fit, then use it as the pattern for the other three.

The full board-by-board cutting layout — which piece comes off which of the six boards — is on the next page.

Cutting Layout

Buy 7 boards (6 get cut, 1 is insurance)

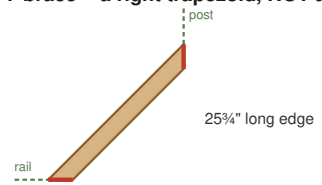
Kayak Cradles — Cutting Layout (both cradles)

Old Town Sportsman 106 PDL · all parts 2×4 (actual 1½" × 3½")

BUY: 6 × 8-ft (96") 2×4

Stud-grade SPF/DF is fine for indoor/garage storage.
Ground-contact pressure-treated or cedar if it lives outdoors.

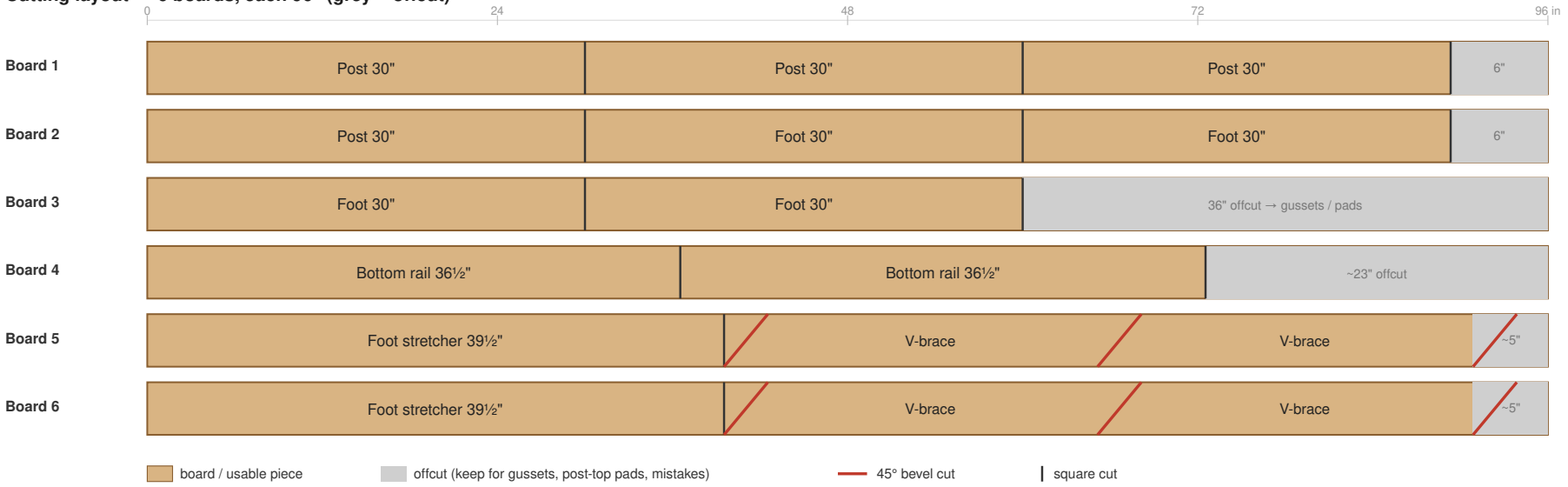
V-brace = a right trapezoid, NOT a parallelogram



Both cuts are 45° to the board's length, but PERPENDICULAR to each other — one horizontal (seats on the rail), one vertical (flush on the post).

Cut one end, FLIP the board end-for-end, then cut the other end at the same 45° — unflipped gives a parallelogram, which won't sit flush on both.

Cutting layout — 6 boards, each 96" (grey = offcut)



Notes

- 16 pieces ≈ 42 board-feet; 6 boards give ~48 ft. This is one way to split them — any grouping that fits 96" works.
- V-brace: set the saw to 45°, cut one end, FLIP the board end-for-end, cut the other end — 25¾" apart on the long edge. It's a right trapezoid (perpendicular cuts), not a parallelogram.
- Cut the square pieces first, then fit one V-brace against a built frame and trim to suit; use it as the pattern for the other 3.
- Feet are not centred — set each one 1" inboard so its outer edge is flush with the post & rail (it overhangs 2" on the inside).
- Put any crown (the slight bow in a board) UP on the feet and the rail.
- Assembly order: feet flat on the ground, 38" on centre → rail screwed to the feet → posts stood up on the feet (flush with the outer edge) → V-braces → stretcher → sling.

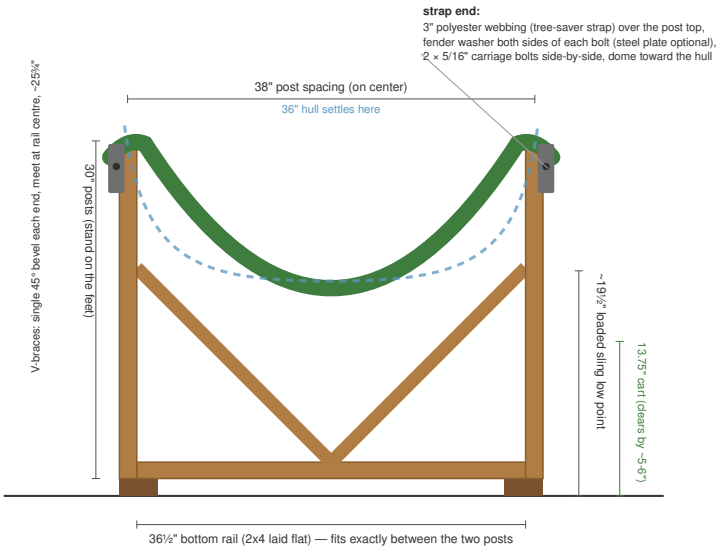
Six boards, cut this way. Grey = offcut, kept for gussets / pads.

Reference Drawings

End view (left) and plan view (right) of one cradle

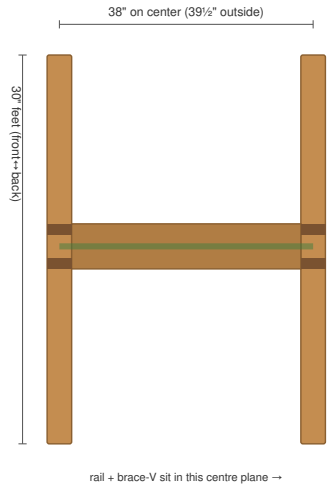
ONE CRADLE — END VIEW (final dimensions)

build 2 · 2x4 throughout · posts dead vertical · braces to the rail center



ONE CRADLE — PLAN VIEW (from above)

shows the two feet — they run front↔back, the only thing resisting fore/aft tip



Build 2. Set them ~6.5 ft (78") apart along the hull.
Round over the top corners of every foot end so you don't stub a toe.
Optional: a stretcher board joining the two feet for extra rack resistance.

Final geometry for the 36" / 107 lb Old Town Sportsman 106 PDL (drive removed). Full cut list, hardware & sling sourcing in BUILD.md.

Key dimensions

Post spacing	38" on center (36 1/2" clear — the 36" hull passes between the posts)
Post length	30" (posts stand on the feet; tops ~31 1/2" off the ground)
Bottom rail	36 1/2" (laid flat; cut to fit exactly between the two posts)
V-brace	~25 3/4" long edge, 45-degree bevel each end (perpendicular, not parallel)
Feet	30" long, one under each post, outer edge flush with the post
Foot stretcher	39 1/2" (optional)
Loaded sling low point	~19 1/2" off the ground (clears a 13.75" cart by ~5-6")
Cradle-to-cradle spacing	~6.5 ft (78") along the hull

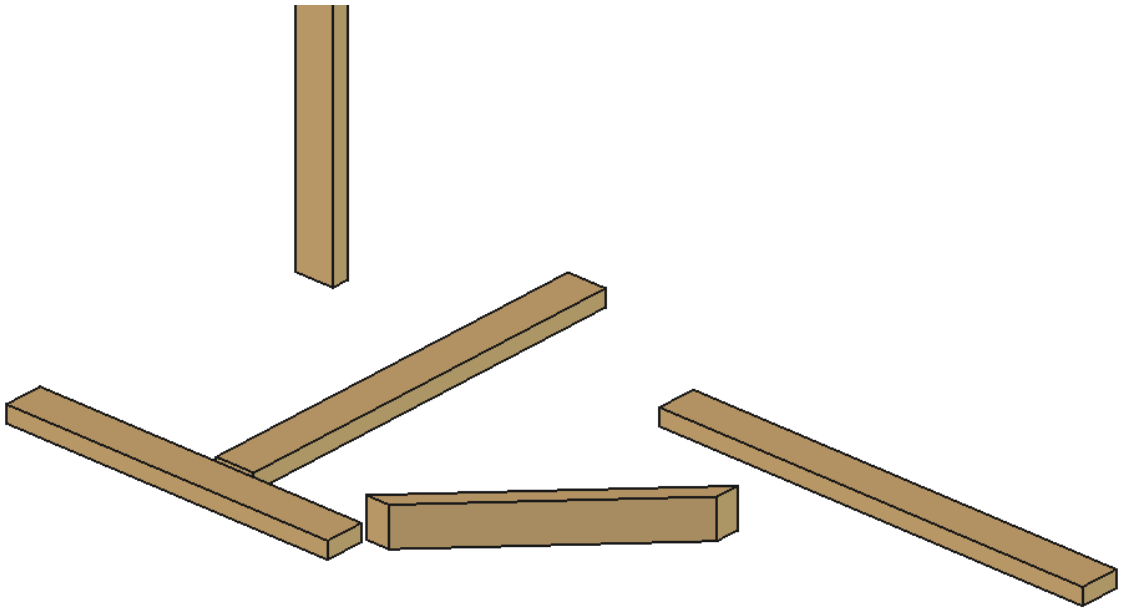
Assembly — Steps 1 & 2

The parts, and the rail screwed to the two feet

Build one complete cradle, then repeat every step for the second. Pre-drill a pilot for every screw within 2" of an end.

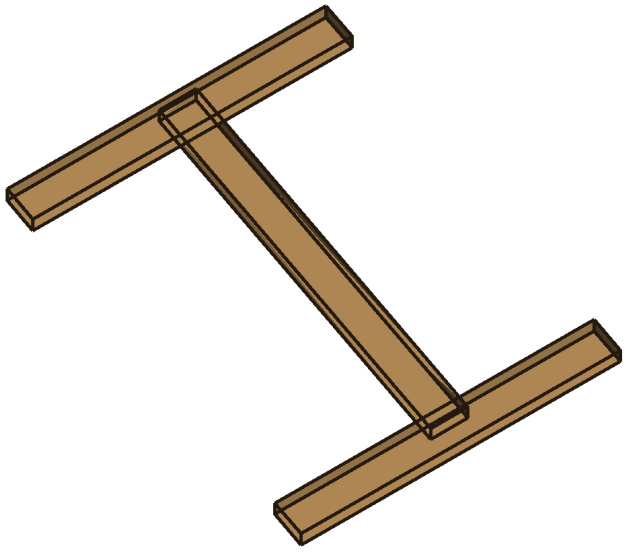
1 Cut the parts

- Per cradle: 2 posts (30"), 1 rail (36 1/2"), 2 feet (30"), + optional 1 stretcher (39 1/2"). The rail must equal the gap between the posts.
- Leave the 2 V-braces as ~28" blanks — you cut them to fit in step 4.



2 Drop in the rail

- Lay both feet flat on the ground, front-to-back, 38" on center — the spacing the posts will stand at — inboard overhangs facing each other.
- Lay the bottom rail flat across the feet, resting on those overhangs, ends 36 1/2" apart (right where the posts' inner faces will land).
- From on top of each rail end drive 2 screws down into the foot — 1" either side of centre (2" apart, 3/4" edge); same layout at both ends.



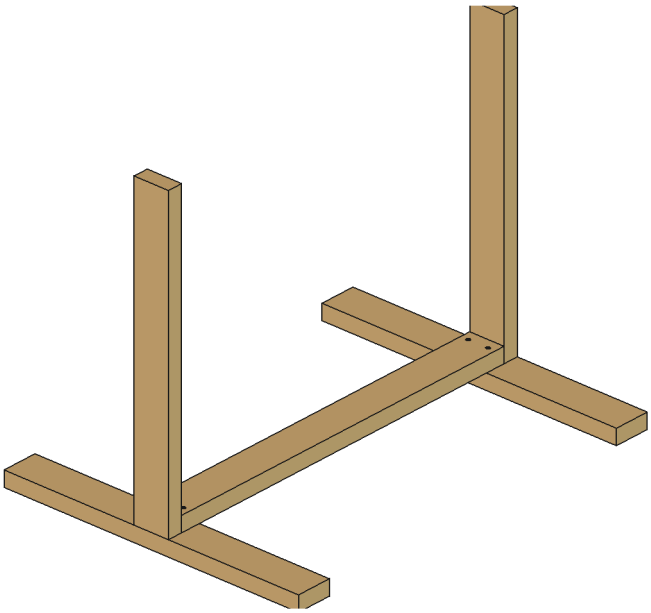
The rail laid across the feet, over their inboard overhang.

Assembly — Steps 3 & 4

Standing the posts, and the V-brace

3 Post + foot sub-assemblies

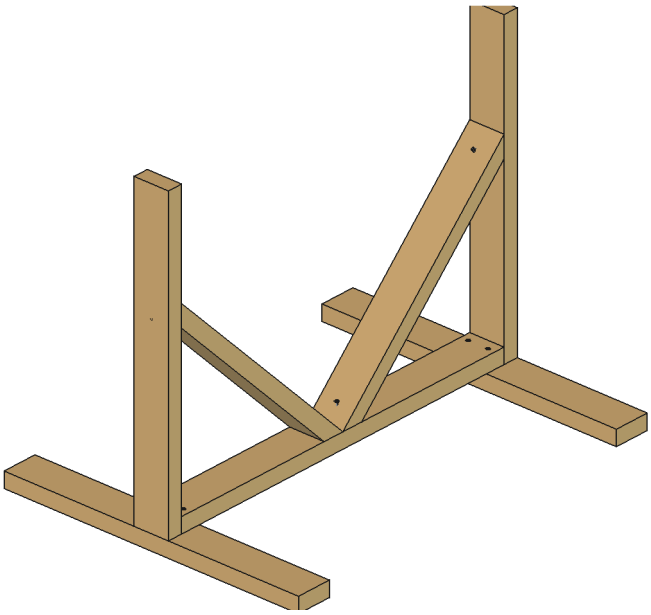
- Stand a post on the outer portion of each foot — post's OUTER face flush with the foot's outer edge. Check the frame is square (equal diagonals) before fastening.
- 2 screws up through the foot into the post — 1" either side of centre (2" apart, 3/4" edge). Same on both.



Frame complete — both posts standing on the rail-equipped feet.

4 Fit the V-braces

- Cut each brace as a right trapezoid, not a parallelogram — 45 deg on one end, then FLIP the board end-for-end and cut the other end at 45 too (unflipped = a parallelogram that won't fit). ~25 3/4" on the long edge. Cut one, test, trim, copy.
- Run each from the CENTRE of the rail top up to a post; the two meet at a point. Bottom bevel flat on rail, top bevel on the post.
- 1 screw down through the brace's bottom end into the rail; 1 screw across from the brace face into the post, ~1 1/2" below the tip.



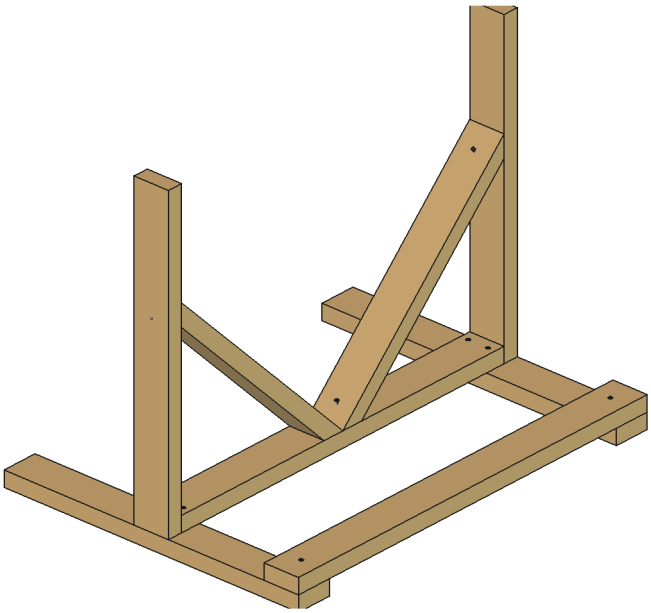
Both braces in; they meet at the rail centre.

Assembly — Steps 5 & 6

Stiffening and finishing the frame

5 Foot stretcher (optional)

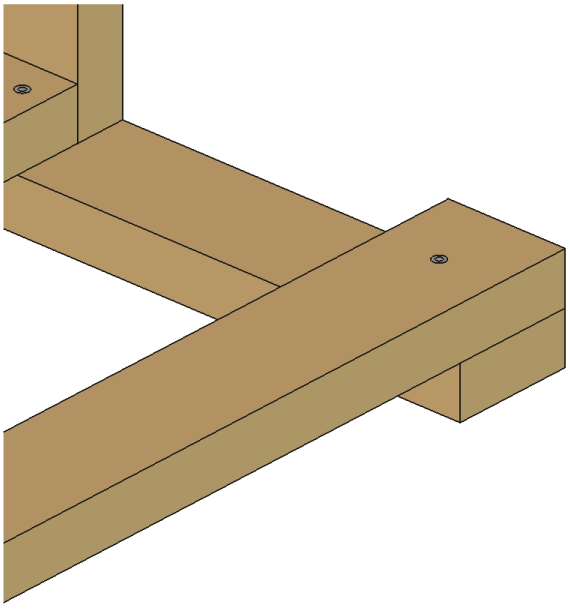
- Lay the 39 1/2" stretcher flat across the tops of both feet near one end.
- 1 screw down through it into each foot. It resists fore/aft racking — worth doing if the cradle will get bumped.



Frame complete: posts, rail, V-brace, stretcher.

6 Ease the foot ends

- Round the top OUTER corners of all four foot ends with a rasp or sander — no stubbed toes, no splintered end grain.
- Good moment to knock the sharp edge off the post tops too.



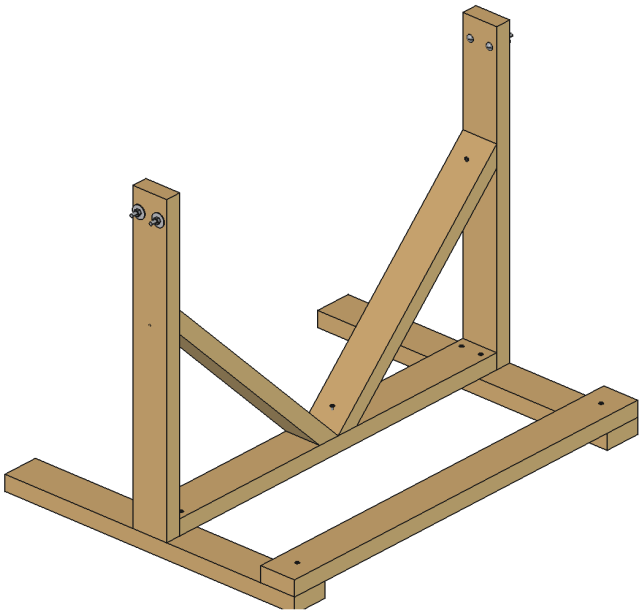
Round the exposed top corners of every foot end.

Assembly — Steps 7 & 8

The pair, and the post-top pads

7 Build the second cradle

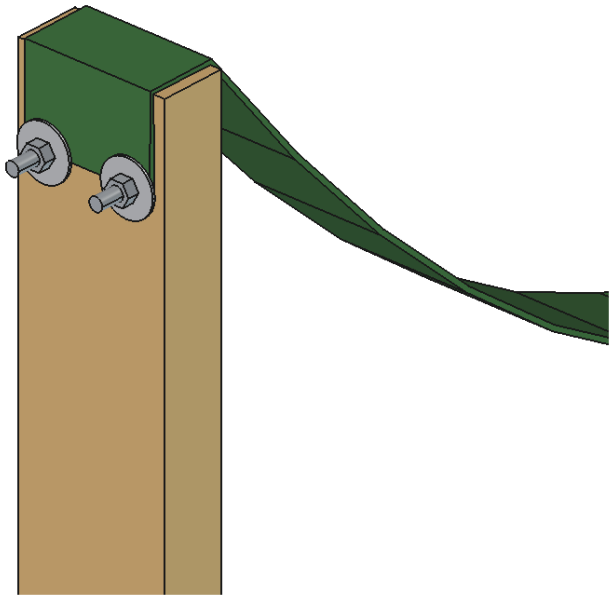
- Repeat steps 1-6. The two cradles must match closely so the boat sits level — use the first one's parts as templates.
- Carriage bolts go in with the sling (next page), but you can drill their holes now.



One finished cradle — build two.

8 Post-top pads (optional)

- Staple a carpet scrap or stick a rubber pad to the top of each post where the hull could touch if the strap goes slack.
- Then fit the sling (next page).



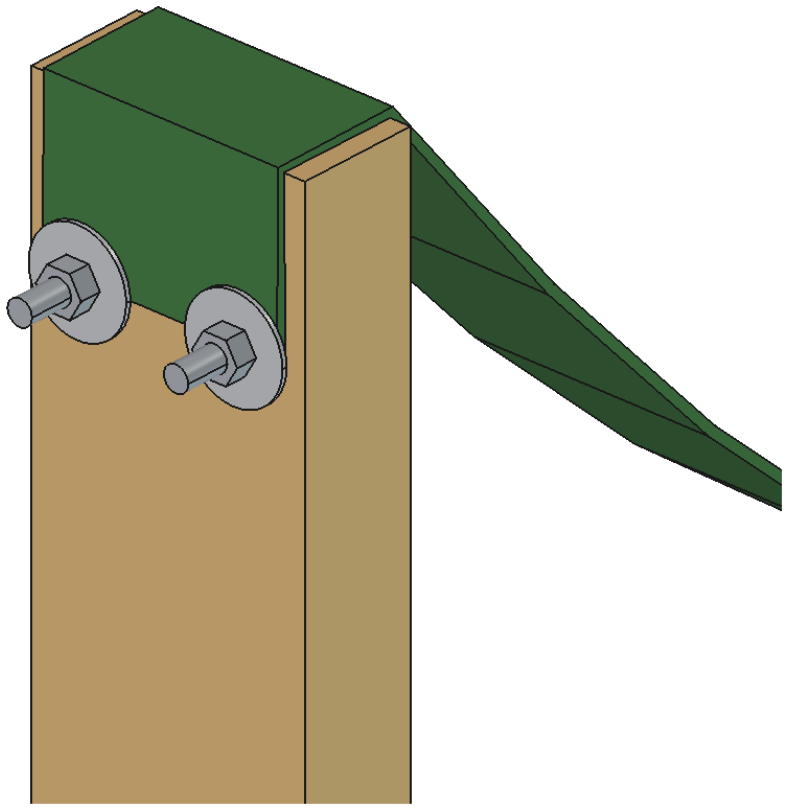
Post top: 2 carriage bolts, fender washer + lock nut on the outside.

The Sling, Setup & Care

Attaching the strap and loading the boat

Fit the sling (do this on each cradle)

1. Cut a length of webbing ~30" longer than the straight-line distance between the two post tops (about 7 ft). Melt-seal both cut ends.
2. Round over the top edge of each post where the strap will bend. Wrap one end over the post top and down the OUTSIDE face about 5".
3. Mark two hole centres side-by-side across the strap, ~2" apart, about 1 1/2-2" down from the post top. (If using the steel plate, lay it on and mark through it.)
4. Drill 5/16" straight through: webbing -> post (-> plate).
5. Push the carriage bolts in FROM THE INSIDE face of the post and tap so the square shoulder bites the wood -- the dome head seats straight against the wood, no washer needed there, and it's the smoothest thing you can put facing the hull. On the outside: fender washer (and plate, if used), then a nylon-lock nut. Snug – firm, not crushing the webbing flat.
6. Do the other post. Before final tightening, pull the strap through so that with the boat in it the low point sits ~19-20" off the ground (~12" of sag). Trim the tail, re-seal.



The strap-end stack: dome head / post / webbing / fender washer / lock nut.

Set it up

Stand the two cradles ~6.5 ft (78") apart on a level floor. With a helper, lower the (drive-removed) kayak in belly-down. Nudge the cradles and adjust strap tension until the hull beds evenly in both slings and sits level.

Care

Keep the webbing out of direct sun when you can — UV is what eventually kills polyester. Re-snug the bolts after the first few weeks as the wood settles. Store the boat with the pedal drive removed so nothing hangs on the strap.

Bill of Materials

Complete shopping + cut summary for two cradles

Everything below is for the PAIR. Full detail in BILL-OF-MATERIALS.md.

Lumber

	Item	Buy
A1	2x4 stud, 8 ft (PT or cedar if outdoors)	7

Cut pieces (2x4; total for both cradles)

	Piece	Length	Qty
B1	Post	30"	4
B2	Foot	30"	4
B3	Bottom rail	36 1/2"	2
B4	V-brace (45 deg each end, perpendicular)	~25 3/4"	4
B5	Foot stretcher (optional)	39 1/2"	2
B6	Brace gusset (optional, from scrap)	~7"	4-6

Fasteners & hardware

	Item	Used	Buy
C1	5/16"-18 x 2 1/2" carriage bolt, zinc	8	10
C2	5/16"-18 nylon-insert lock nut, zinc	8	10
C3	5/16" fender washer (1 1/4-1 1/2" OD)	8	10
C4	3" #9-#10 exterior deck screw	28	1 lb box
C5	Tie plate ~3x5" galv. (optional)	0 / 4	0 / 4

Screw schedule (item C4)

Joint	Per cradle	Pair
Foot -> post (up through foot)	4	8
Rail end -> foot (down through rail)	4	8
Brace -> post (horizontal into post)	2	4
Brace -> rail (up through rail)	2	4
Stretcher -> foot (optional)	2	4
TOTAL	14	28

Sling

	Item	Qty
D1	2" or 3" polyester tow / tree-saver strap	~14 ft

One 2" x 20 ft polyester tow strap (Husky, HD #59925) cut in half does both cradles. Cut the loops/hooks off, melt-seal the ends. Polyester, not nylon.

Cost, the pair (rough 2026 big-box): 7 x 2x4 ~$28 · bolts/nuts/washers ~$10 · 1 lb screws ~$9 · 2" x 20 ft strap ~$15 = about $55-75. Other files: BUILD.md (full text), cut-sheet.png, photo-match-design.png, cradle_model.FCStd/.step, cradle_freecad.py (parametric — change POST_SPACING etc. and re-run).